Sagy Ephrati

Curriculum vitae

Personal details

Name:	Sagy Richard Ephrati
Nationality:	Dutch
Email:	sagy@chalmers.se
Address:	Department of Mathematical Sciences, Chalmers University of Technology and University of Gothenburg, 41296, Gothenburg, Sweden

Appointments

Postdoctoral Researcher, Chalmers University of Technology — Dec 2023 - Present
Gothenburg, Sweden

Researcher, University of Twente — Sep 2023 - Nov 2023
Enschede, The Netherlands

Education

PhD in Applied Mathematics, University of Twente — Sep 2019 - Sep 2023
Thesis: *Data-driven Stochastic Modeling for Coarsened Computational Geophysical Fluid Dynamics* — Enschede, The Netherlands
Supervisor: Prof. dr. ir. Bernard J. Geurts

MSc in Applied Mathematics, University of Twente — Sep 2017 - Jul 2019
Thesis: *Direct numerical simulation of bubble-wall interaction* — Enschede, The Netherlands
Supervisor: Prof. dr. ir. Bernard J. Geurts

Internship, Technische Universität Dresden — Sep 2018 - Mar 2019
Assignment: *Comparison of numerical methods for multiphase flows* — Dresden, Germany
Supervisor: Prof. Jochen Froehlich

BSc in Applied Mathematics, University of Twente — Sep 2013 - Jul 2017
Thesis: *A numerical study of the Ecovat with IFISS* — Enschede, The Netherlands
Supervisor: Prof. dr. ir. Bernard J. Geurts

Teaching experience

At the University of Twente

Numerical Mathematics, Bachelor Applied Mathematics	Fall 2022
Numerical Mathematics, Bachelor Applied Mathematics	Fall 2021
Numerical Mathematics, Bachelor Applied Mathematics	Fall 2020
Student Assistant, Linear Algebra, Bachelor Applied Mathematics	Fall 2017
Student Assistant, Linear Algebra, Bachelor Applied Mathematics	Fall 2016
Student Assistant, Linear Algebra, Bachelor Applied Mathematics	Fall 2015
Student Assistant, Linear Algebra, Bachelor Applied Mathematics	Fall 2014
Student Assistant, Calculus, various study programmes	2014-2019

Scholarships

Resestipendium, Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond	Summer 2025
Resestipendium, Stiftelsen för Vetenskaplig Forskning och Utbildning i Matematik (SVeFUM)	Summer 2025
Scholarship Mathematics 2024, The Royal Swedish Academy of Sciences (KVA)	Fall 2024
Resestipendium, The Royal Society of Arts and Sciences in Gothenburg (KVVS)	Summer 2024
Workshop organisation scholarship, Stiftelsen för Vetenskaplig Forskning och Utbildning i Matematik (SVeFUM) *Joint scholarship with Erik Jansson, Klas Modin, Michael Roop*	Summer 2024

[1]Updated July 10, 2025

Organized activities

Workshop co-organizer — Oct 2024
Four-day workshop on geometric methods and stochastic reduction for fluid models — Kristineberg, Sweden

Workshop co-organizer — May 2023
Five-day workshop on geometric methods and stochastic reduction for fluid models — Enschede, The Netherlands

Publications

Preprints

4. **Thermal quasi-geostrophic model on the sphere: derivation and structure-preserving simulation** (joint with M. Roop). (2025).
3. **Critical latitude in global quasi-geostrophic flow on a rotating sphere** (joint with A.D. Franken, E. Luesink and B.J. Geurts). (2024).
2. **An exponential map free implicit midpoint method for stochastic Lie-Poisson systems** (joint with E. Jansson, A. Lang and E. Luesink). (2024).
1. **Geometric derivation and structure-preserving simulation of quasi-geostrophy on the sphere** (joint with E. Luesink, A.D. Franken and B.J. Geurts). (2024).

Peer-reviewed articles

10. **Probabilistic data-driven turbulence closure modeling by assimilating statistics**. J. Comp. Phys. (2025).
9. **On spectral scaling laws for averaged turbulence on the sphere** (joint with E. Jansson and K. Modin). Physica D (2025).
8. **Casimir preserving numerical method for global multilayer geostrophic turbulence** (joint with A.D. Franken, E. Luesink and B.J. Geurts). J. Comp. Phys. (2025).
7. **Two minimal-variable symplectic integrators for stochastic spin systems** (joint with E. Jansson). Phys. Rev. E. (2025). (links)
6. **Continuous data assimilation closure for modeling statistically steady turbulence in large-eddy simulation** (joint with A.D. Franken, E. Luesink, P. Cifani and B.J. Geurts). Phys. Rev. Fluids. (2025). (links)
5. **Casimir preserving stochastic Lie–Poisson integrators** (joint with E. Luesink, P. Cifani and B.J. Geurts). Adv. Contin. Discrete Models (2024). (links)
4. **Data-driven spectral turbulence modeling for Rayleigh-Bénard convection** (joint with P. Cifani and B.J. Geurts). J. Fluid Mech. (2023). (links)
3. **Data-driven stochastic spectral modeling for coarsening of the two-dimensional Euler equations on the sphere (2023)** (joint with P. Cifani, M. Viviani and B.J. Geurts). Phys. Fluids (2023). (links)
2. **Data-driven stochastic Lie transport modelling of the 2D Euler equations** (joint with P. Cifani, E. Luesink and B.J. Geuts). J. Adv. Model. Earth Syst. (2023). (links)
1. **Computational modeling for high-fidelity coarsening of shallow water equations based on subgrid data** (joint with E. Luesink, G. Wimmer, P. Cifani and B.J. Geurts). Multiscale Model. Simul. (2022). (links)

Refereed proceedings

2. **Stochastic data-driven POD-based modeling for high-fidelity coarsening of two-dimensional Rayleigh-Bénard turbulence** (joint with P. Cifani and B.J. Geurts). Direct and Large Eddy Simulation XIII: Proceedings of DLES13 (2023). (links)
1. **Sparse-stochastic model reduction for 2D Euler equations** (joint with P. Cifani and M. Viviani). Stochastic Transport in Upper Ocean Dynamics II (STUOD 2022) (2023). (links)

Presentations

2025

Efficient stochastic closure modelling by assimilating statistics. PDEs on the Sphere, São Paulo, Brazil, May 2025.
Error-correcting stochastic models based on ideal large-eddy simulation. Stochastic Transport in Upper Ocean Dynamics (STUOD) meeting, Imperial College, London, United Kingdom, Feb 2025.
An exponential map-free integrator for stochastic Lie–Poisson systems. BIT '65, Uppsala, Sweden, Jan 2025.

2024

On stochastic computational geophysical fluid dynamics: geometric integration and data-driven modeling. Seminar for Numerical Methods in Plasma Physics, Max Planck Institute for Plasma Physics, Garching bei München, Germany, Nov 2024.
Low-dimensional stochastic closure models by assimilating statistics. Workshop on geometric methods and stochastic reduction for fluid models, Kristineberg, Sweden, Oct 2024.
Data assimilation closure for low-dimensional fluid models. ECCOMAS, Lisbon, Portugal, Jun 2024.
Data assimilation closure for LES of Rayleigh-Bénard convection. Direct and Large Eddy Simulation (DLES) 14, Erlangen, Germany, Apr 2024.

2023

Stochastic modeling for coarse computational geophysical fluid dynamics. Seminar on Computational and Applied Mathematics, Chalmers University of Technology, Gothenburg, Sweden, Sep 2023.
Data-driven spectral modeling for the quasi-geostrophic equations. PDEs on the Sphere, Grenoble, France, Jul 2023.
Data-driven stochastic forcing for uncertainty quantification and subgrid-scale modeling in geophysical fluid dynamics. Seminar for Machine Learning and UQ in Scientific Computing, Centrum Wiskunde & Informatica (CWI), Amsterdam,The Netherlands, May 2023.
Data-driven spectral modeling for the quasi-geostrophic equations. 22nd Computational Fluids Conference, Cannes, France, Apr 2023.

2022

Data-driven POD-based modeling for high-fidelity coarsening of two-dimensional Rayleigh-Bénard turbulence. Direct and Large Eddy Simulation (DLES) 13, Udine, Italy, Oct 2022.
Structure-preserving data-driven modeling for the two-dimensional Euler equations. Workshop on geometric methods and stochastic reduction for fluid models, L'Aquila, Italy, Sep 2022.
Structure-preserving POD-based forcing for the two-dimensional Euler equations. SciCADE, Reykjavik, Iceland, Jul 2022.
Structure-preserving POD-based forcing for the two-dimensional Euler equations. ECCOMAS, Oslo, Norway, Jun 2022.
Extracting and employing dynamic structures for reduced-order modeling. Stochastic Transport in Upper Ocean Dynamics (STUOD) Sandbox Workshop, online, Feb 2022.

2021

Assessment of High-Resolution Numerical Simulation of Bubble-Wall Interaction. 13th International ERCOFTAC Symposium on Engineering Turbulence Modelling and Measurements, online, Sep 2021.

Research visits

Imperial College, London, United Kingdom	Feb 2025
Gran Sasso Science Institute, L'Aquila, Italy	Sep 2022
Gran Sasso Science Institute, L'Aquila, Italy	Sep 2021
Imperial College, London, United Kingdom	Oct 2019

Other activities

Participant, Summer School on Bayesian Filtering: fundamental theory and numerical methods at ICMS, Bayes Centre, Edinburgh, United Kingdom	May 2024

Referee activity

Physics of Fluids, IMA Journal of Numerical Analysis, Journal of Computational Dynamics

Outreach

Twente Pre-U, hosting one-day mathematical workshops for high school students	2014-2019

Skills

Programming

Matlab, Python

Languages

Dutch	Native speaker
English	Fluent
German	Good working knowledge
Swedish	Good working knowledge
Hebrew	Basic knowledge